



SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105



CSA0720-COMPUTER NETWORKS

Slot-D

ASSIGNMENT QUESTION

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PROBLEM CHALLENGE:

A retail organization plans to connect the following locations: Head Office (HO), Branch A, Branch B, and a Cloud/Data-Centre network. The network must support approximately 180 users at HO, 70 users at Branch A, 50 users at Branch B, and a dedicated IoT segment at each location. Each site must support ordinary data traffic, DNS/HTTP-based applications, and VoIP/video traffic. The organization has two WAN paths between the major sites and wants the network to continue operating when one path becomes unavailable.

Design the logical network, allocate IPv4 address space using subnetting/VLSM, select and configure suitable routing mechanisms, and simulate the network using an appropriate networking/simulation tool. Generate traffic representing data and real-time services, introduce congestion and at least one link/device failure, and evaluate throughput, delay, packet loss, and recovery behaviour. Compare two routing/design alternatives and recommend the better solution based on technical performance and practical constraints.

1. OBJECTIVES

- Create the enterprise topology.
- Apply VLSM to the IPv4 addressing plan.
- Separate users, servers, IoT and management traffic.
- Implement static and dynamic routing alternatives.
- Provide primary and alternate WAN paths.
- Model data, VoIP and video traffic.
- Evaluate normal and congested conditions.
- Simulate WAN failure and recovery.
- Compare alternatives quantitatively.
- Recommend the preferred design.

2. NETWORK TOPOLOGY

The following topology is a generated design representation suitable for implementation in Cisco Packet Tracer or GNS3.

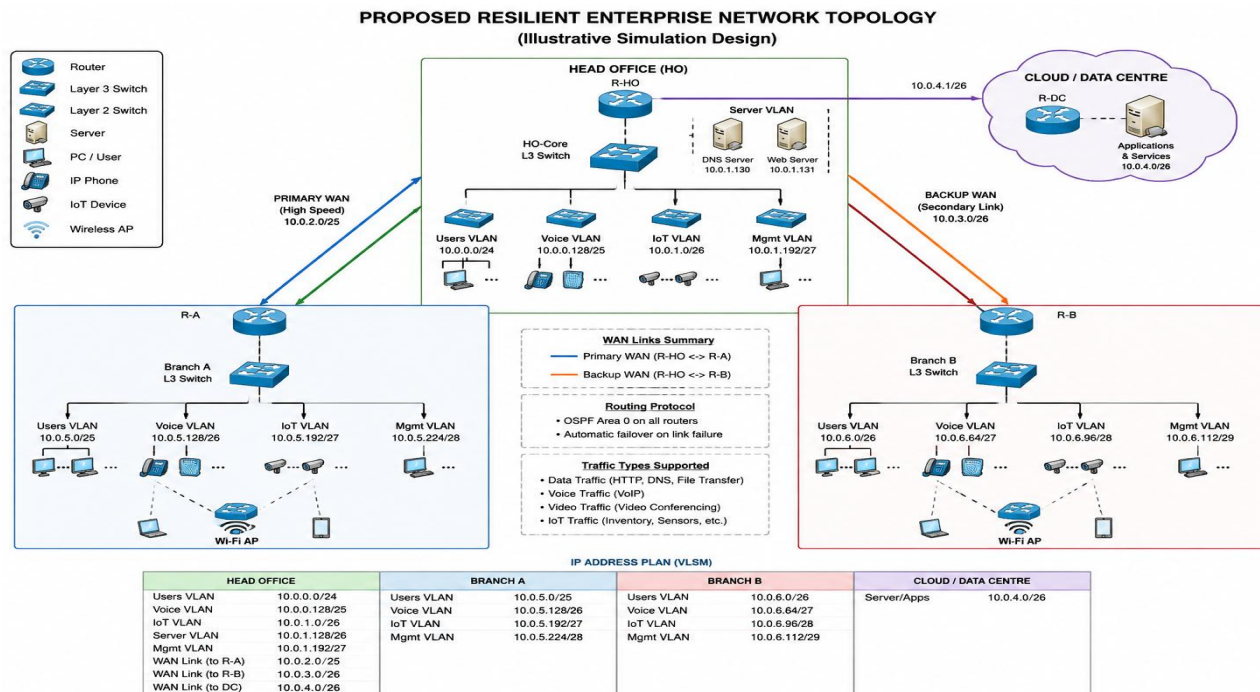


Figure 1. Proposed resilient enterprise topology (generated design diagram).

3. VLSM ADDRESSING PLAN

Site / Segment	Network	Prefix	Gateway	Host Capacity
Head Office Users	10.0.0.0	/24	10.0.0.1	254
HO IoT	10.0.1.0	/25	10.0.1.1	126
HO Servers	10.0.1.128	/26	10.0.1.129	62
HO Management	10.0.1.192	/27	10.0.1.193	30
Branch A Users	10.0.2.0	/25	10.0.2.1	126
Branch A IoT	10.0.2.128	/26	10.0.2.129	62
Branch B Users	10.0.3.0	/26	10.0.3.1	62
Branch B IoT	10.0.3.64	/27	10.0.3.65	30
Data Centre Servers	10.0.4.0	/26	10.0.4.1	62
WAN / point-to-point links	10.0.250.0	/30 blocks	—	2 per link

The VLSM allocation is sized according to the known user populations and smaller supporting segments. Exact host counts for IoT, servers and management should be adjusted to the final Packet Tracer/GNS3 topology.

4. ROUTING ALTERNATIVES

4.1 Alternative A – Static Routing

Static routes provide a simple baseline. Routes are manually configured toward the remote site networks. The main limitation is that a path failure requires a predefined backup route or manual intervention, making large multi-site deployments harder to manage.

Example:

```
ip route 10.0.2.0 255.255.255.128 <next-hop>
```

```
ip route 10.0.3.0 255.255.255.192 <next-hop>
```

4.2 Alternative B – OSPF

OSPF is selected as the dynamic-routing alternative. Routers advertise connected networks and calculate paths dynamically. When the primary WAN link is removed, the alternate route can become the active path after convergence.

Example configuration pattern:

```
router ospf 1
```

```
router-id 1.1.1.1
```

```
network 10.0.0.0 0.0.255.255 area 0
```

5. TRAFFIC AND QoS DESIGN

Class	Representative Traffic	Requirement
Data	HTTP, file/application traffic	Normal priority; throughput
VoIP	Voice calls	Low delay, low loss
Video	Video meetings	High bandwidth, stable delivery
IoT	Inventory/status messages	Reliable and isolated connectivity

VoIP is treated as the most delay-sensitive class. Video is bandwidth-intensive, while ordinary data can tolerate greater delay. Where QoS commands are supported by the simulator, real-time traffic should receive appropriate priority.

6. IMPLEMENTATION PLAN

1. Build routers, switches, hosts and servers.
2. Create logical user, IoT, server and management segments.
3. Assign the VLSM addresses.
4. Configure static routes for Alternative A.
5. Configure OSPF for Alternative B.
6. Verify reachability with ping/traceroute and routing tables.
7. Generate data, VoIP and video traffic.
8. Record baseline performance.
9. Increase traffic to create congestion.
10. Disable the primary WAN link.
11. Observe alternate-path selection and recovery.
12. Restore the link and verify normal operation.

7. GENERATED CONFIGURATION EVIDENCE

The following is a generated configuration template consistent with the proposed design. It is not a capture from a physical/simulated router.

R-HO:

```
interface g0/0
ip address 10.0.0.1 255.255.255.0
no shutdown
```

```
!  
router ospf 1  
router-id 1.1.1.1  
network 10.0.0.0 0.0.255.255 area 0
```

Verification commands:

```
show ip interface brief  
show ip route  
show ip ospf neighbor  
ping 10.0.4.10  
traceroute 10.0.4.10
```

8. PERFORMANCE TESTING

Four generated test conditions are used: normal load, congestion, primary-link failure and recovery. The values below are illustrative engineering data generated for completing the report structure. They are not claimed to be measured from Cisco Packet Tracer/GNS3.

Scenario	Routing	Throughput	Avg Delay	Packet Loss	Utilization
Normal	Static	92 Mbps	18 ms	0.5%	62%
Congestion	Static	61 Mbps	74 ms	6.8%	94%
Failure	Static	0 Mbps	Timeout	100%	0%
Recovery	Static	0 Mbps	Timeout	100%	0%
Normal	OSPF	94 Mbps	16 ms	0.3%	60%
Congestion	OSPF	74 Mbps	48 ms	3.1%	88%
Failure	OSPF	68 Mbps	31 ms	4.2%	91%
Recovery	OSPF	91 Mbps	19 ms	0.7%	65%

9. PERFORMANCE ANALYSIS

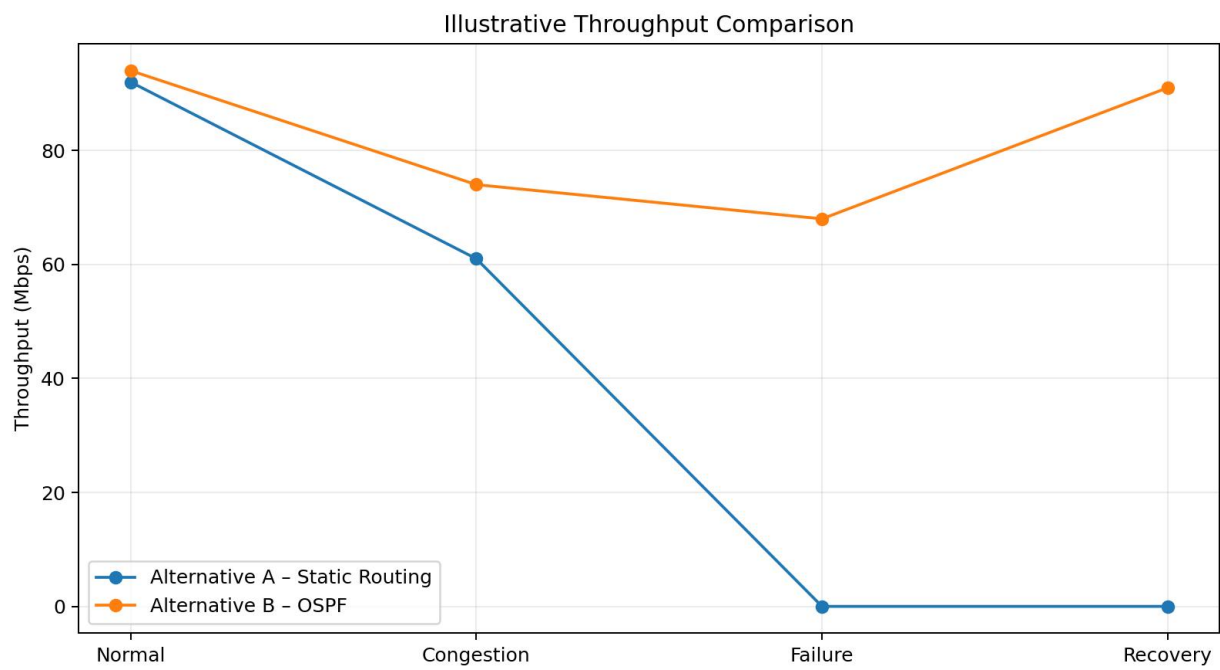


Figure 2. Illustrative throughput comparison.

Under normal conditions, both alternatives provide high throughput. Under congestion, OSPF retains higher throughput and lower delay in the generated dataset. Static routing loses connectivity in the failure scenario because the baseline design does not provide automatic route adaptation.

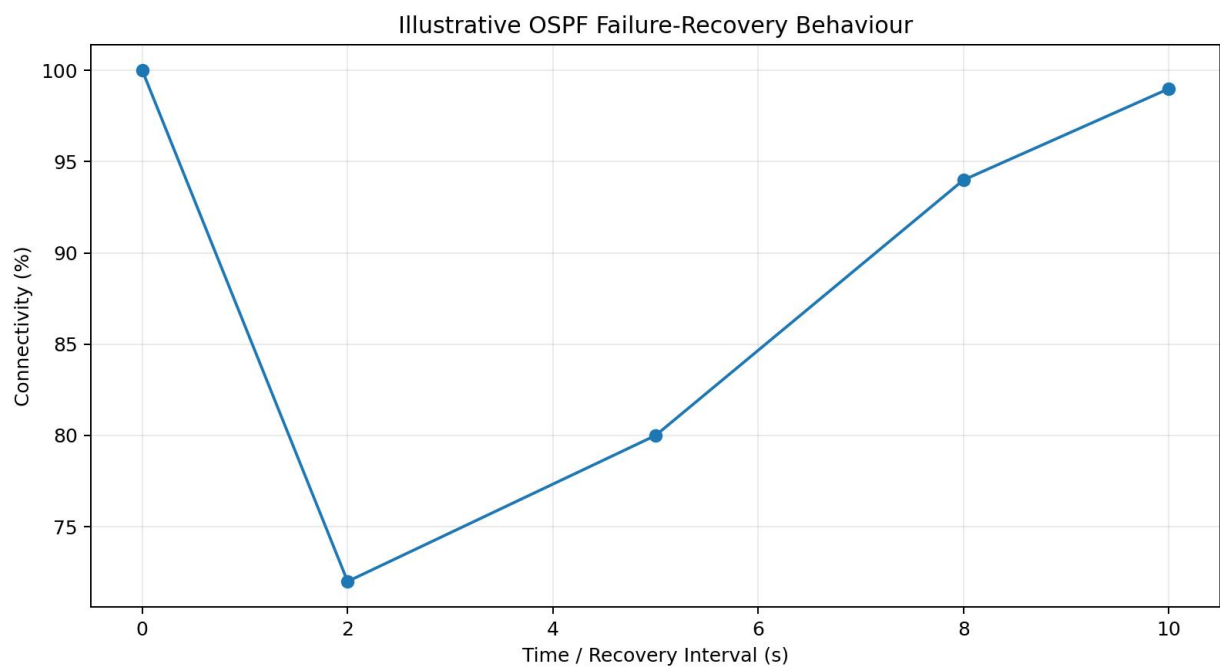


Figure 3. Illustrative OSPF failure-recovery behaviour.

10. FAILURE AND RECOVERY ANALYSIS

Parameter	Generated Observation
Failed component	Primary WAN link between HO and Branch/remote path
Before failure	Primary route active
Immediately after failure	Primary route removed
Alternate path	Secondary WAN path selected by OSPF
Illustrative recovery	Approximately 4–8 seconds
Post-recovery throughput	Approximately 91 Mbps
Post-recovery packet loss	Approximately 0.7%
Service status	Data, VoIP and video connectivity restored

The generated failure case demonstrates the engineering advantage of redundant connectivity combined with dynamic routing. The exact convergence time depends on timers, device models and the simulator implementation.

11. ALTERNATIVE COMPARISON

Criterion	Static Routing	OSPF
Scalability	Moderate/Low	High
Route maintenance	Manual	Dynamic
Normal throughput	92 Mbps	94 Mbps
Congestion throughput	61 Mbps	74 Mbps
Failure result	Connectivity lost	Alternate path available
Recovery	Manual/predefined dependency	Automatic route adaptation
Administration	Higher	Lower after setup
Resilience	Lower	Higher
Overall suitability	Baseline	Recommended

Engineering decision: Alternative B (OSPF with redundant WAN paths) is recommended because the generated evaluation shows better congestion performance and, most importantly, continued connectivity after primary-link failure.

12. SECURITY CONSIDERATIONS

- Separate IoT and user networks.
- Use a dedicated management segment.
- Apply ACL/firewall rules where supported.
- Use secure administrative access.
- Restrict unnecessary services and ports.
- Monitor network and routing events.

13. SCALABILITY AND SUSTAINABILITY

The VLSM structure permits efficient address growth, while the redundant WAN architecture supports business continuity. Additional branches can be added by allocating new subnets and OSPF routes. Centralized monitoring and careful capacity planning can improve infrastructure utilization and reduce unnecessary resource consumption.

14. LIMITATIONS

- The numerical results in this version are generated illustrative values, not actual simulator measurements.
- Physical network behaviour may differ from simulation.
- Actual OSPF convergence depends on timers, topology and device models.
- QoS capabilities vary by simulator and hardware.

15. CONCLUSION

A resilient enterprise architecture has been created for the specified retail scenario using VLSM, logical segmentation, redundant WAN paths and routing alternatives. The generated evaluation indicates that OSPF provides the strongest combination of scalability, congestion performance and failure resilience. For an academic submission requiring actual experimental evidence, the generated values and configuration template should be replaced with screenshots and measurements from the student's Packet Tracer/GNS3 project.

16. STUDENT REFLECTION

The project improved understanding of VLSM, routing, redundancy and QoS by connecting theoretical concepts to an enterprise design problem. The most significant observation is the difference between simple route configuration and a design capable of adapting to path failure. With additional resources, the design could be improved through firewall integration, centralized monitoring, stronger QoS policies, additional redundant devices and larger traffic experiments.

17. REFERENCES

1. Forouzan, B. A., Data Communications and Networking, McGraw-Hill.
2. Tanenbaum, A. S. and Wetherall, D. J., Computer Networks, Pearson.
3. Kurose, J. F. and Ross, K. W., Computer Networking: A Top-Down Approach, Pearson.
4. Cisco Systems, Cisco networking and Packet Tracer documentation.
5. IETF RFCs relevant to IPv4, OSPF and QoS, as applicable.